



Mid Term Exam
Winter Semester: 2026

Paper Information			
Artificial Intelligence/ Advanced Artificial Intelligence			
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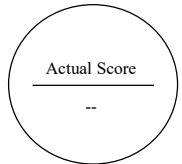
Exam Instructions
- Fill the above 'Student Details' section carefully and complete. - Cheating is strictly prohibited and action will be taken.

Marking Scheme	
Question No.	Actual Score / Max. Score
Q1	/20
Q2	/20
Q3	/60
Total Score:	/ 100

Learning Outcomes Achievement		
Learning Outcome	Question #	Student's Achievement

Q1) Color with red **ONE** correct answer (letter) for each of the following MCQs in the table below. [20 Marks]

1	2	3	4	5	6	7	8	9	10
a	a	a	a	a	a	a	a	a	a
b	b	b	b	b	b	b	b	b	b
c	c	c	c	c	c	c	c	c	c
d	d	d	d	d	d	d	d	d	d



- Which prompt engineering technique is used when you ask the AI to respond as a professional or expert?.
 - Clarifying context
 - Role-play
 - Data labeling
 - Fine-tuning
- What is the main purpose of clarifying context in prompt engineering?
 - To reduce the length of the question
 - To provide relevant details and scope for better answers
 - To confuse the AI intentionally
 - To remove examples from the prompt
- What does providing references mean in prompt engineering?
 - Asking the model to search the internet
 - Giving the model trusted information to use in its answer
 - Limiting the model to short answers
 - Removing examples from the prompt
- What is the main purpose of using symbols in prompt engineering?
 - To decorate the prompt
 - To shorten the input text
 - To clearly identify parts of the text that require specific tasks.
 - To train the AI model



5. Which of the following best describes formal reasoning?
 - a. Contextual and subjective thinking
 - b. Human-like understanding
 - c. Symbolic logic and mathematics
 - d. Creative storytelling
6. Natural Language Reasoning is also known as:
 - a. Logical reasoning
 - b. Mathematical reasoning
 - c. Informal reasoning
 - d. Statistical reasoning
7. Which characteristic best fits natural language (informal) reasoning?
 - a. Strict rules and symbols
 - b. Exact mathematical formulas
 - c. Human-like and contextual understanding
 - d. No ambiguity
8. Which type of reasoning relies more on context and nuance?
 - a. Formal reasoning
 - b. Symbolic reasoning
 - c. Natural language reasoning
 - d. Mathematical reasoning
9. Which of the following is NOT a benefit of Chain of Thought prompting ?
 - a. Improves accuracy
 - b. Enhances explainability
 - c. Helps with debugging
 - d. Reduces model size
10. Large Reasoning Models are mainly linked to which reasoning type?
 - a. System 1
 - b. Hybrid emotional reasoning
 - c. System 2 (natural language-based)
 - d. No reasoning

Q2) For each question, color True (T) or False (F) as appropriate.

1	Natural language processing (NLP) is a field of artificial intelligence that allows computers to perceive and understand the world around them, just like a human, using cameras, data, and algorithms to replicate vision	T	F
2.	A way for computers to learn to recognize patterns from many examples and to make decisions, without being directly programmed with specific rules for every decision, is referred to as data structures.	T	F
3	Generative AI models are mainly designed to classify data rather than create new content	T	F
4	DALL-E 2 model is mainly used to generate sound and music.	T	F
5	LLaMA 2 is an image generation model.	T	F
6	Jukebox is a generative AI model used to generate sound or music.	T	F
7.	ChatGPT is trained using reinforcement learning from human feedback (RLHF), which combines reinforcement learning techniques, such as rewards and comparisons, with human guidance and feedback	T	F
8.	Autonomous mobile robots (AMRs) are robots equipped with advanced sensors and navigation systems that can perform tasks autonomously, such as maintaining quality control and inventory management, reducing labor costs, and increasing production efficiency.	T	F
9.	Generative AI is a type of AI that can create new text compositions, realistic images, high-quality video, original audio, or programming code. It employs generative models that discover the rules and patterns of their input training data and then create new data that resembles and builds on the original content.	T	F
10.	Copilot takes into account the surrounding text (the context) and the overall topic when generating suggestions.	T	F

[20 Marks]

Q3)

- A) Large Language Models can be guided using different prompt engineering strategies.

Consider the following prompting techniques: Zero-Shot Prompting, Role Prompting
Chain-of-Thought Prompting, Self-Refine Prompting, Least-to-Most Prompting

Question: Compare these five prompting techniques.

1. For each prompting technique, provide a short definition in your own words.
2. Discuss the strengths and limitations of each strategy.
3. Describe when each technique would be most appropriate.
4. Provide a concrete example prompt for each method.

Your answer should demonstrate a clear understanding of how prompting strategies influence reasoning quality and output structure.

Solution:

Zero Shot Prompting

Definition: Zero shot prompting is probably the most used prompting by an average user of LLMs. Zero shot prompting means prompting an AI with no context, examples, or relevant information regarding the topic and solely relying on information found in its pretrained dataset.

Strengths: Zero shot prompting is very useful in test cases to see how intuitive the AI is on a subject or to test its tolerance to ambiguity. It is also a great way to obtain quick answers that won't require much validation. It can also be a great way to start a conversation with an AI to start to organize your thoughts before prompting the AI in a more formal fashion.

Limitations: Due to the lack of context around the question and the broadness of the AI's dataset, the answers retrieved may be very vague or include information that is not entirely accurate or even relevant to the topic at hand. Information may also be outdated because the AI are trained at certain intervals and not continuously up to date on the latest information and data.

Example: "Give me some thought starts around a new coding project."

Role Prompting

Definition: Role Prompting is the act of instructing the AI to take on the role of a character. The AI returns responses in the tone and style the character would return.

Strengths: Role Prompting can help anchor the AI in the domain of the relevant data, for example prompting an AI to be a finance analyst when speaking to futures predictions will return results more tailored to the topic than if you prompted it to be a Carpenter.

Limitations: Role prompting has been proven to generate a significantly higher rate of hallucinations than other prompting methods. This is because the model will make things up to fit the narrative of the persona. Also, role-prompting presents a false sense of security. Information that is false can be really convincing when said in the “voice” of an expert. Lastly, because prompting helps to anchor AI, it could also lead to perpetuation of stereotypes or biases it has in its own dataset for the topic rather than bringing in more data from other areas.

Example: “You are a Project Manager. You have been tasked with presenting the plants portfolio to the plant manager’s boss next week. Given the information attached, put together a high-level summary of all projects and develop a speech that is relevant to the given audience.

Chain of Thought Prompting

Definition: Chain of Thought prompting involves prompting an AI to solve a problem or think about a question in multiple steps. This can be done by providing similar problems to the one at hand and having it derive the solution to the problem at hand by pulling from the similarities, it can be explicit step by step instructions on how to solve the problem, or it can simply be prompting the AI to engage in “step by step” thinking as well. It helps the AI to think of the problem as a sequence of solutions and not just one size fits all.

Strengths: This prompting method has proven very effective in mathematics, strategy, and other areas that require multi step solutions. It can also be used to verify the integrity of the solution as well as all steps in the process can be interrogated.

Limitations: Chain of thought increases calculation time and overhead as opposed to other prompting. Because of the resulting multi step process, the amount of resources allocated is increased directly correlational to the amount of steps.

Example: Automatic CoT : “I am trying to Budget for my home improvement project. I am looking to add a couple walls, add some cabinets, install a sink, and add some lighting. Let’s think about the project step by step so I can find out how much money I will need in total and when I will need certain amounts of it.”

Self Refine Prompting

Definition: Self Refine prompting is having the model analyze it's own output and refine it's answer that it gave. It forces the model to engage in an iterative process with itself to improve upon it's own answers.

Strengths: Self refining has shown to produce better results on a lot of benchmark subjects such as writing and mathematics. The answers tend returned at the end of self refining tend to be more palatable to humans. It allows the AI to improve upon itself without training.

Limitations: The AI may return an answer that sounds better but only reinforces incorrect information in an even more appealing context. In situations where the AI just plain makes errors it is impossible for the AI to refine it's own error as it can't see it. Self Refine Prompting also has diminishing returns, it cannot just continuously refine itself to be infinitely good, so although the output may be different it might not be objectively better.

Example: Initial Prompt: "Write me a paper for my AI class "

Feedback Prompt: "Go through the paper you just wrote and check to make sure all subjects are explained well and clearly. Give precise feedback as to how each area of the paper could be improved"

Refinement Prompt: "Rewrite the paper you wrote initially but make sure to include all the feedback you found in your answer to the previous prompt."

Least to Most Prompting

Definition: Least to Most prompting is breaking up larger problems into smaller sub problems. Answers to smaller subproblems can be appended to the new prompt with the next step in the sub problem or the prompt can explicitly explain the task and its relevant subproblems.

Strengths: Less memory intensive solving more smaller problems rather than one large problem. It produces better results in similarly benchmarked tasks than CoT.

Limitations: Incorrect answers at one step of the problem translate to the rest of the subproblems creating compounding errors.

Example:

"Problem: Solve $2+2+6$

Step 1: $2+2$

Step 2: {Step 1 Result} + 6

Answer: {Step 2 Result}"

[20 Marks]

B) Describe the key components of a Retrieval-Augmented Generation (RAG) system and explain the role of each component in the overall workflow. Then, illustrate your answer with a concrete application example of a RAG system. In your example, explain:

- The type of knowledge source used for retrieval,
- How relevant information is retrieved and integrated into the generation process.
- Explain why a Retrieval-Augmented Generation (RAG) approach is more suitable than a standalone large language model for this application.

Solution:

The first step in RAG is to gather knowledge sources to be fed into the system as “Tried and True” information for the system to reference. The first step is to embed these documents in a vector database. Each item in the knowledge source is broken into individual chunks called vectors. These vectors are stored in a vector database. These vectors are mathematical representations of each chunk of the knowledge source. Next, the user's prompt or query is converted into vectors using the same exact conversion process that the knowledge sources underwent. After that, the RAG system performs calculations to see which parts of the knowledge base are most similar to the query. I personally have had experience with cosine similarity in which the direction of the vectors are compared to each other to determine if they are similar or not. If it is found that there are vectors in the vector database that are similar to the query, the system will pull the corresponding Knowledge Source information into the LLM's context window. The LLM now can use that information to answer the query, drastically cutting down on hallucinations.

In my own personal endeavors I have recently developed a system very similar to RAG but with a little twist that I believe makes it more reliable and the outputs more verifiable. Although it isn't traditional RAG, I believe the concept is very similar and hope that it will serve well for answering this question.

At work (General Motors) we store TONS of information in SQL databases. These databases are dense, full of information ranging from finance to geometric information. Ultimately after digging around in there for a while I began to realize how inaccessible this information was. I began to think of doing a RAG type system that would embed our database and allow users to Query the databases with Natural Language. I quickly ran into issues with LLM's because although you can use RAG, these databases are so large and complex that hallucinations are rampant and even



worse a hallucination is almost unverifiable because of the complex SQL you would have to generate to verify.

With this in mind I thought:

Why not embed my Database Schema into a bot that has been trained in SQL prompts? Now instead of having to carry around the weight of the giant database all the Bot has to do is carry around the weight of what it already knows about SQL just tuned to my database. Next, I can pass the generated query to the database and return the result. Now with the result I can hand that off a regular LLM and generate a summary in natural language.

Not only does this solve the problem of having to store this entire database in vector space and risking hallucinations by similar values, it makes everything verifiable from a checking aspect because now I could literally see the code that went into the query to answer question. All the while being totally accessible to those that have never even seen the databases or know what SQL is.

[20 Marks]

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C)

1. Explain the difference between a **base LLM** and a **fine-tuned LLM**. Then, provide **one example** where fine-tuning improves the model's response compared to a base model.
2. Discuss the main benefits of fine-tuning.

A Base LLM is sort of like an off the shelf LLM. It has pretrained weights on whatever data source the creator used to train it on. When prompted, the Base LLM tokenizes the query and creates vectors out of the tokens. These vectors are high dimensional objects, and once created they are passes through the LLM's Transformer. The transformer has multiple layers of heads with self attention followed by feedforward layers. In a Base LLM these objects inside of the Transformer have the "Off the Shelf" pretrained weight values. It uses it's "broad knowledge" to analyze the prompt and produce the next most statistically likely token in it's response.

Fine Tuning, such as LoRA Fine Tuning, is the process in which projection matrices created from the new knowledge source are inserted into very specific spots in the transformer to alter the output of a layer before it is passed on to the next layer. This allows training the Base LLM on data outside of it's initial training source, directly changing the way it behaves, without having to retrain the entire model.

I am working on implementing a good example of LoRA fine tuning at work. In a previous question I had talked about a SQL generator that interrogates a database. Currently I gave the LLM my database schema in it's prompt so it can reference it when generating queries. Soon I am going to fine tune the model to my database so it insert lower level matrices at critical points in the Base LLM to fine tune it to my specific application.

[20 Marks]

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**End of Examination... Good Luck!**

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